

THIRUVENKADAPRASATH.S

192312016

10	Write a Python Program to Implement Expectation & Maximization Algorithm
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CSV FILE

Value

1.2

1.5

1.7

1.1

1.9

1.4

6.8

7.1

6.9

7.5

7.2

6.6

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
def gaussian(x, mean, var):
```

```
    return (1 / np.sqrt(2 * np.pi * var)) * np.exp(  
        -((x - mean) ** 2) / (2 * var)  
    )
```

```

df = pd.read_csv("em_algorithm_data.csv")
X = df["Value"].values
n_samples = len(X)

m1, m2 = 2.0, 5.0
v1, v2 = 1.0, 1.0
w1, w2 = 0.5, 0.5

iterations = 20

for i in range(iterations):
    pdf1 = gaussian(X, m1, v1)
    pdf2 = gaussian(X, m2, v2)

    gamma1 = (w1 * pdf1) / (w1 * pdf1 + w2 * pdf2)
    gamma2 = (w2 * pdf2) / (w1 * pdf1 + w2 * pdf2)

    n1 = np.sum(gamma1)
    n2 = np.sum(gamma2)

    m1 = np.sum(gamma1 * X) / n1
    m2 = np.sum(gamma2 * X) / n2

    v1 = np.sum(gamma1 * (X - m1) ** 2) / n1
    v2 = np.sum(gamma2 * (X - m2) ** 2) / n2

    w1 = n1 / n_samples
    w2 = n2 / n_samples

print("--- Estimated Parameters after Convergence ---")

```

```
print(f"Cluster 1 -> Mean: {m1:.4f}, Variance: {v1:.4f}, Weight: {w1:.4f}")
```

```
print(f"Cluster 2 -> Mean: {m2:.4f}, Variance: {v2:.4f}, Weight: {w2:.4f}")
```

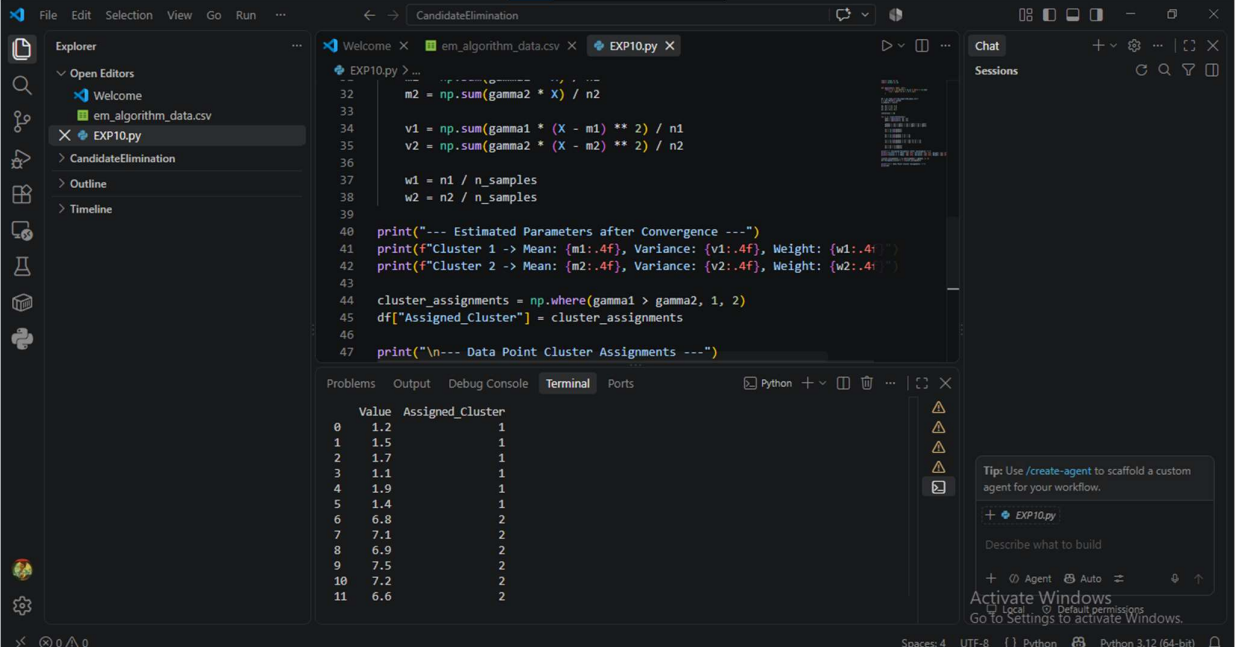
```
cluster_assignments = np.where(gamma1 > gamma2, 1, 2)
```

```
df["Assigned_Cluster"] = cluster_assignments
```

```
print("\n--- Data Point Cluster Assignments ---")
```

```
print(df)
```

OUTPUT



The screenshot shows a VS Code editor with a Python script named `EXP10.py` and its output in the terminal. The script calculates the EM algorithm's convergence parameters for two clusters. The terminal output displays the estimated parameters and the cluster assignments for 12 data points.

```
32 m2 = np.sum(gamma2 * X) / n2
33
34 v1 = np.sum(gamma1 * (X - m1) ** 2) / n1
35 v2 = np.sum(gamma2 * (X - m2) ** 2) / n2
36
37 w1 = n1 / n_samples
38 w2 = n2 / n_samples
39
40 print("--- Estimated Parameters after Convergence ---")
41 print(f"Cluster 1 -> Mean: {m1:.4f}, Variance: {v1:.4f}, Weight: {w1:.4f}")
42 print(f"Cluster 2 -> Mean: {m2:.4f}, Variance: {v2:.4f}, Weight: {w2:.4f}")
43
44 cluster_assignments = np.where(gamma1 > gamma2, 1, 2)
45 df["Assigned_Cluster"] = cluster_assignments
46
47 print("\n--- Data Point Cluster Assignments ---")
```

	Value	Assigned_Cluster
0	1.2	1
1	1.5	1
2	1.7	1
3	1.1	1
4	1.9	1
5	1.4	1
6	6.8	2
7	7.1	2
8	6.9	2
9	7.5	2
10	7.2	2
11	6.6	2